



Mining User Opinions to Support Requirement Engineering: An Empirical Study

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App Reviews as a Resource:

Rich source of user opinions to understand requirements, preferences, and perceptions.

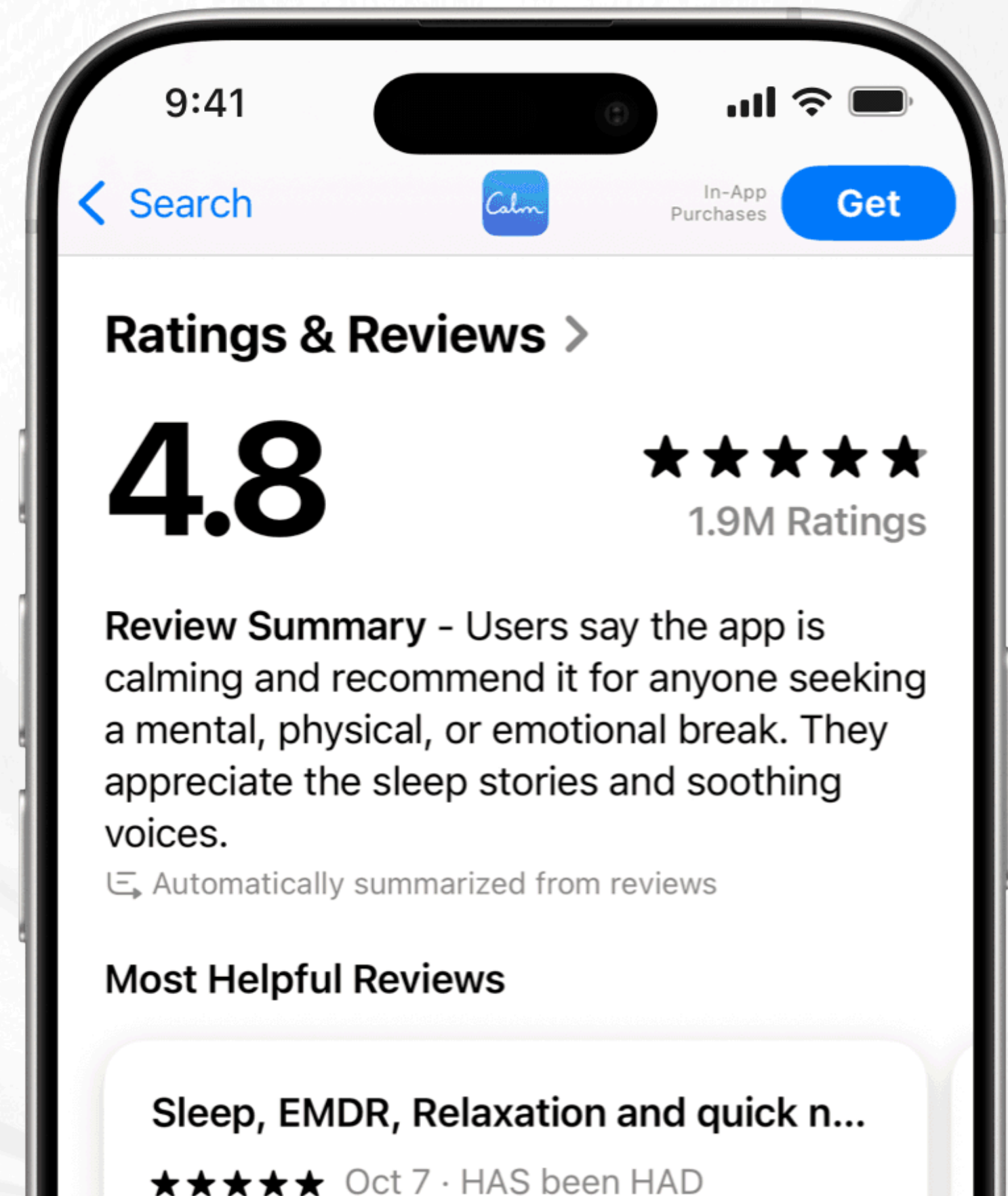


Challenges:

- Large quantity of reviews (**thousands/day**).
- **Noisy content** obstructs engineering activities.

Research Gap:

Difficulty replicating studies and comparing approaches due to inconsistent methods/datasets



Key Concepts and Contributors



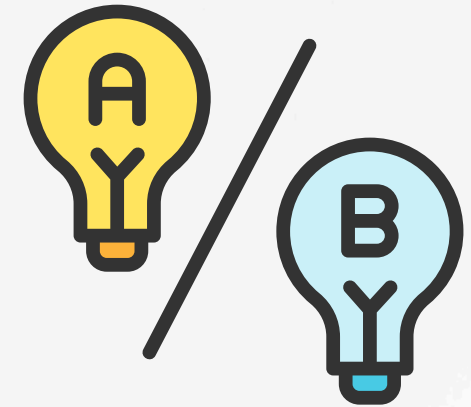
Problems with Opinion Mining in App Reviews



Reviews are noisy, unstructured, and high in volume, making manual analysis difficult.



Past studies are hard to replicate because they used inconsistent datasets and methods.



No standardized ground truth exists, limiting fair comparison of different approaches.

Key Concepts and Contributors

Methods

New annotated dataset



Collected 1,000 app reviews across 8 apps, manually labeled into 1,521 feature–sentiment pairs for reliable evaluation.

Evaluated



GuMa – word patterns + sentiment tool.
SAFE – linguistic patterns (features only).
ReUS – dependency parsing (features + sentiment).

Evaluation metrics



Performance measured using precision (accuracy of extractions) and recall (coverage of correct extractions).

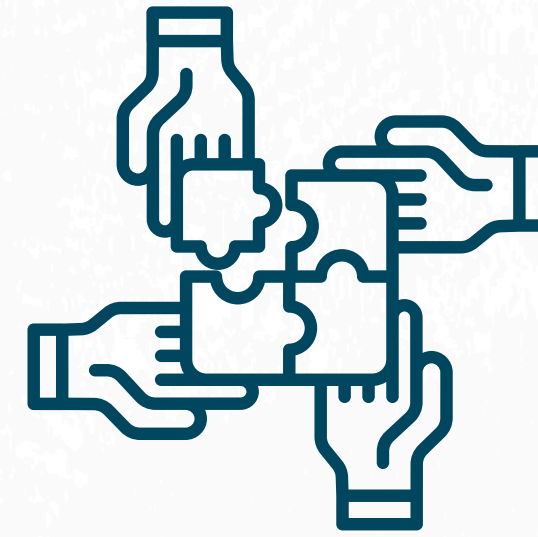
Critical Analysis

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Findings

- Feature extraction weak (e.g., SAFE recall 0.34 vs 0.71).
- Sentiment: fair for positive/neutral, poor for negative.
- GuMa $\approx$ ReUS despite different methods.



Contributions

- Rigorous empirical comparison.
- Public benchmark dataset.
- Evidence tools not practical yet.

Critical Analysis

07



Strengths

- Clear annotation + reliability measures.
- Public dataset → reproducibility.
- Multiple approaches, side-by-side.
- Explicit feature-matching rules.



Limitations

- Dataset small vs millions of reviews.
- Manual annotation → subjective.
- Misses newer ML/deep learning methods.
- ReUS designed for e-commerce, not apps.

Implications & Connections

Practical Impact:

Current tools unreliable, esp. for negative feedback.

For RE:



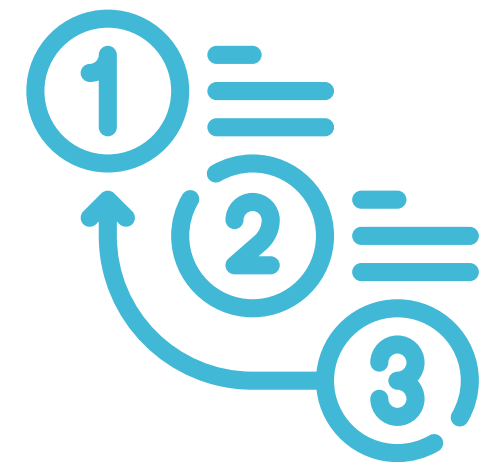
Validation

Check feature acceptance.



Elicitation

Discover new requirements.



Prioritization

Spot top problem areas

Research Value: Benchmark dataset + reproducible methods



THANK YOU